

will always register the value of 4 V irrespective of the position of the slider.

24 Ans: D

At point A, the resultant electric field is pointing towards the left.

At points B and C, the resultant electric field is pointing towards the right.

At point D, the electric field due to $+4Q$ is pointing towards the right and that due to $-Q$ is pointing towards the left. Considering the magnitude of $\frac{Q}{r^2}$ for both charges at point D, it is possible for the resultant electric field to be zero.

25 Ans: D